Ratios

A ratio compares two amounts: for every 3 of these, 2 of those.

3 circles for every 2 squares



3 : 2

Say “3 to 2”. Two groups is 6 : 4, the same ratio.

A ratio table: multiply both rows by the same number

cups of flour	2	4	6	8
eggs	3	6	9	12

× 2, × 3, × 4:
both numbers change together.

Part to part, part to whole

A class has 3 boys for every 5 girls. Boys to girls is 3 : 5.
Out of every 8 kids, 3 are boys. Boys to all kids is 3 : 8.

Your turn!

$$4 : 5 = 12 : \boxed{15}$$

For teachers

Goal: ratio language, equal ratios and ratio tables (6.RP.1, 6.RP.3a). Say “For every 3 of these there are 2 of those.” A ratio table multiplies both rows by the same number.

Ratios: Practice

Both numbers change together.

Part 1 Write the ratio

Count the circles, then the squares.

1) 

circles to squares = 3 : 2

2) 

circles to squares = 4 : 1

3) 

circles to squares = 2 : 5

4) 

circles to squares = 5 : 3

5) 

circles to squares = 1 : 4

6) 

circles to squares = 6 : 2

7) 

circles to squares = 3 : 4

8) 

circles to squares = 2 : 3

9) 

circles to squares = 5 : 2

Part 2 Ratio tables

Multiply both rows by the same number.

10)

cups	2	4	6	8	10
people	3	6	9	12	15

11)

laps	3	6	12	18	30
minutes	4	8	16	24	40

12)

red	5	15	20	35	45
blue	2	6	8	14	18

Ratios: Practice (continued)

Part 2 Ratio tables (continued)

13)

pens	4	8	20	32	40
dollars	3	6	15	24	30

14)

cats	1	3	5	6	12
dogs	6	18	30	36	72

15)

boys	7	14	21	42	70
girls	5	10	15	30	50

Part 3 Equal ratios No help this time.

16) $2 : 3 = 8 : 12$

17) $5 : 4 = 15 : 12$

18) $1 : 6 = 5 : 30$

19) $3 : 7 = 6 : 14$

20) $4 : 9 = 12 : 27$

21) $6 : 5 = 24 : 20$

22) $2 : 9 = 12 : 54$

23) $7 : 3 = 35 : 15$

24) $8 : 5 = 16 : 10$

Ratios: Practice (continued)

Part 3 Equal ratios (continued)

25) 3 : 10 = 12 : 40

26) 9 : 4 = 27 : 12

27) 5 : 8 = 30 : 48

28) 4 : 7 = 20 : 35

29) 1 : 9 = 8 : 72

30) 6 : 7 = 18 : 21

31) 10 : 3 = 20 : 6

32) 3 : 8 = 21 : 56

33) 2 : 11 = 6 : 33

34) 7 : 4 = 42 : 24

35) 5 : 9 = 20 : 36

36) 8 : 3 = 24 : 9

Ratios: Practice (continued)

Part 3 Equal ratios (continued)

37) $11 : 4 = 22 : \boxed{8}$

38) $3 : 5 = \boxed{27} : 45$

39) $12 : 5 = 24 : \boxed{10}$

Part 4 Simplest form Divide both numbers by the same number.

40) $8 : 12 = \boxed{2} : \boxed{3}$

41) $6 : 10 = \boxed{3} : \boxed{5}$

42) $6 : 24 = \boxed{1} : \boxed{4}$

43) $15 : 6 = \boxed{5} : \boxed{2}$

44) $8 : 14 = \boxed{4} : \boxed{7}$

45) $9 : 24 = \boxed{3} : \boxed{8}$

46) $28 : 8 = \boxed{7} : \boxed{2}$

47) $25 : 30 = \boxed{5} : \boxed{6}$

48) $6 : 27 = \boxed{2} : \boxed{9}$

Ratios: Practice (continued)

Part 4 Simplest form (continued)

49) 12 : 10 = 6 : 5

50) 7 : 21 = 1 : 3

51) 16 : 6 = 8 : 3

Part 5 Part to whole No help this time.

52) 3 boys for every 5 girls: boys to all kids = 3 : 8

53) 2 boys for every 7 girls: boys to all kids = 2 : 9

54) 4 boys for every 3 girls: boys to all kids = 4 : 7

55) 1 boy for every 5 girls: boys to all kids = 1 : 6

56) 5 boys for every 6 girls: boys to all kids = 5 : 11

57) 6 boys for every 1 girl: boys to all kids = 6 : 7

58) 2 boys for every 3 girls: boys to all kids = 2 : 5

59) 7 boys for every 4 girls: boys to all kids = 7 : 11

Ratios: Practice (continued)

Part 6 More equal ratios

60) $2 : 6 = 4 : \boxed{12}$

61) $4 : 8 = \boxed{16} : 32$

62) $2 : 4 = \boxed{18} : 36$

63) $11 : 12 = \boxed{77} : 84$

64) $3 : 5 = 18 : \boxed{30}$

65) $11 : 7 = 77 : \boxed{49}$

66) $7 : 12 = \boxed{49} : 84$

67) $3 : 12 = \boxed{9} : 36$

68) $1 : 2 = 7 : \boxed{14}$

69) $6 : 4 = 48 : \boxed{32}$

70) $12 : 11 = \boxed{60} : 55$

71) $6 : 12 = 24 : \boxed{48}$

Ratios: Practice (continued)

Part 6 More equal ratios (continued)

72) $9 : 3 = 18 : \boxed{6}$

73) $10 : 12 = \boxed{20} : 24$

74) $9 : 4 = 81 : \boxed{36}$

75) $10 : 2 = 30 : \boxed{6}$

76) $2 : 7 = \boxed{8} : 28$

77) $10 : 6 = 20 : \boxed{12}$

78) $5 : 8 = \boxed{40} : 64$

79) $2 : 5 = 12 : \boxed{30}$

80) $4 : 3 = \boxed{20} : 15$

81) $11 : 7 = \boxed{88} : 56$

82) $6 : 3 = 36 : \boxed{18}$

83) $12 : 10 = \boxed{36} : 30$

Ratios: Practice (continued)

Part 7 More simplest form

84) 12 : 32 = 3 : 8

85) 3 : 18 = 1 : 6

86) 4 : 10 = 2 : 5

87) 35 : 40 = 7 : 8

88) 27 : 6 = 9 : 2

89) 14 : 49 = 2 : 7

90) 25 : 15 = 5 : 3

91) 27 : 12 = 9 : 4

92) 32 : 20 = 8 : 5

93) 2 : 4 = 1 : 2

94) 6 : 36 = 1 : 6

95) 4 : 6 = 2 : 3

Ratios: Practice (continued)

Part 8 Remember? Problems from earlier packets.

96) $(6 + 9) \times 4 = 60$

97) $9 \times \frac{2}{7} = \frac{18}{7}$

98) $0.6 \times 0.9 = 0.54$

99) $3{,}472 \div 8 = 434$

100) $\frac{7}{8} - \frac{1}{3} = \frac{13}{24}$

101) $4 \text{ ft } 2 \text{ in} = 50 \text{ in}$

Part 9 Word problems Use a ratio table if it helps.

102) A recipe uses 2 cups of rice for every 3 cups of water. How much water goes with 8 cups of rice?

Number sentence 2 : 3 = 8 : 12

Answer 12 cups

Draw or show your work

103) There are 4 red marbles for every 5 blue ones. There are 20 blue marbles. How many red ones?

Number sentence 4 : 5 = 16 : 20

Answer 16 red marbles

Draw or show your work

104) A team won 6 games and lost 9. What is the ratio of wins to losses in simplest form?

Number sentence 6 : 9 = 2 : 3

Answer 2 : 3

Draw or show your work

Ratios: Practice (continued)

Part 9 Word problems (continued)

105) Paint is mixed 3 cans of blue to 1 can of white. How many cans of blue go with 7 cans of white?

Number sentence 3 : 1 = 21 : 7

Answer 21 cans

Draw or show your work

106) Challenge: A class has 3 boys for every 4 girls. There are 28 kids. How many are girls?

Number sentence 3 + 4 = 7 kids per group, 28 ÷ 7 = 4 groups, 4 × 4 = 16

Answer 16 girls

Draw or show your work

Check-In: Ratios

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Write the ratio

1) 

circles to squares = 4 : 3

Part 2 Ratio tables

2)

cups	3	6	12	18
people	5	10	20	30

Part 3 Equal ratios

3) 3 : 4 = 15 : 20

4) 4 : 3 = 24 : 18

5) 5 : 7 = 10 : 14

Part 4 Simplest form

6) 8 : 28 = 2 : 7

7) 10 : 6 = 5 : 3